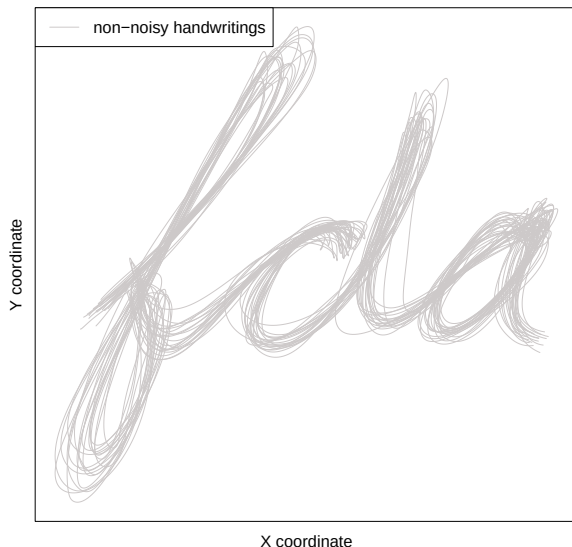


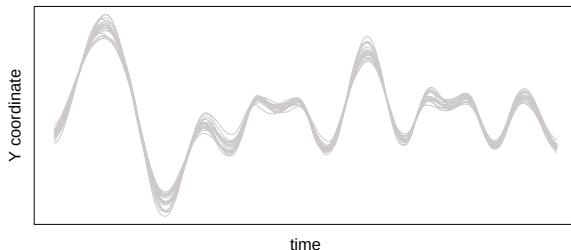
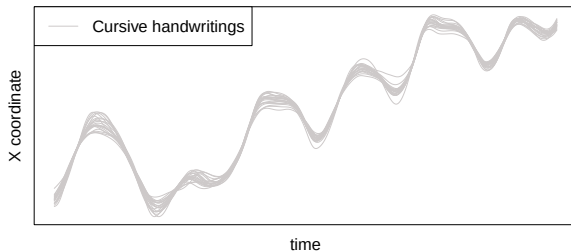
Motivating Example: Cursive Handwriting

fda – Cursive handwriting samples

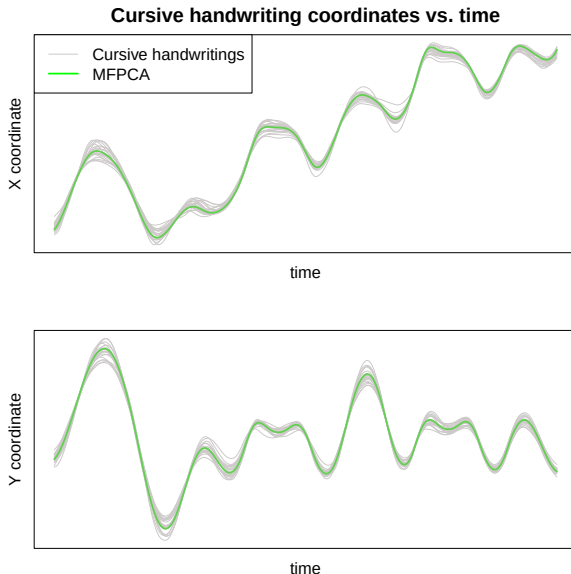


Motivating Example: Cursive Handwriting

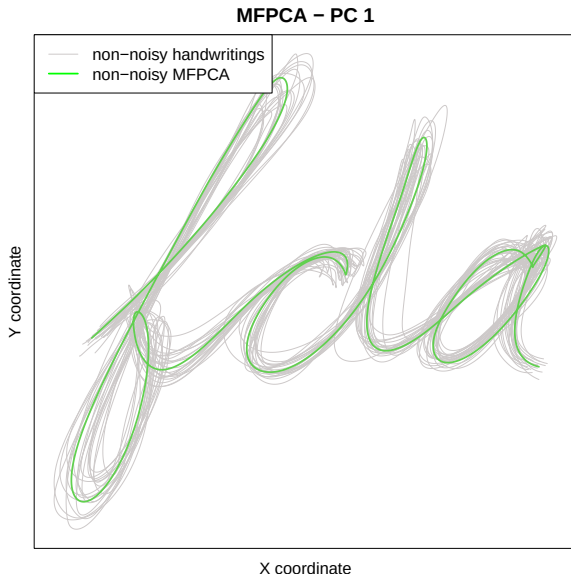
Cursive handwriting coordinates vs. time



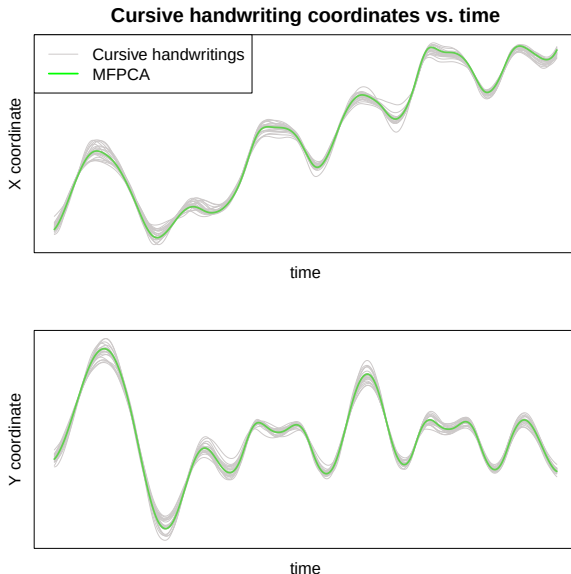
Motivating Example: Cursive Handwriting



Motivating Example: Cursive Handwriting

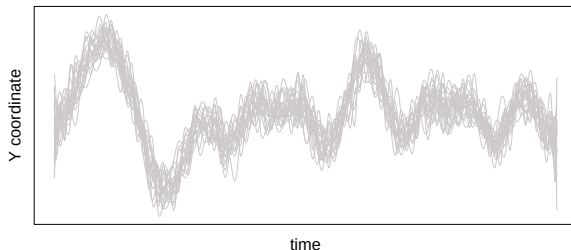
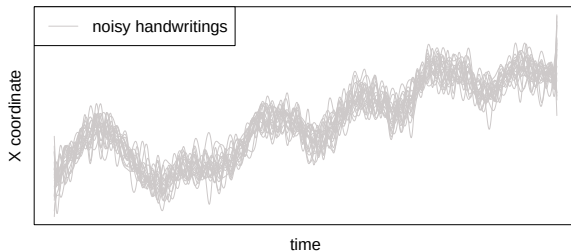


Motivating Example: Cursive Handwriting



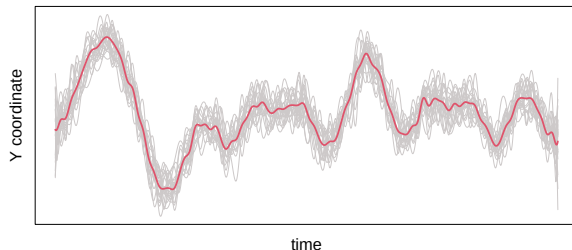
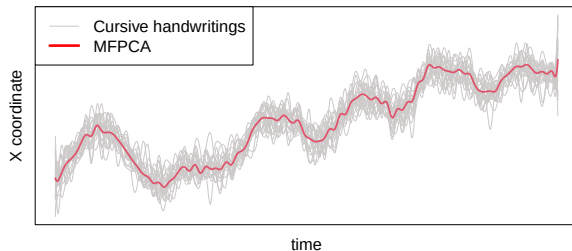
Motivating Example: Cursive Handwriting

noisy handwriting coordinates vs. time



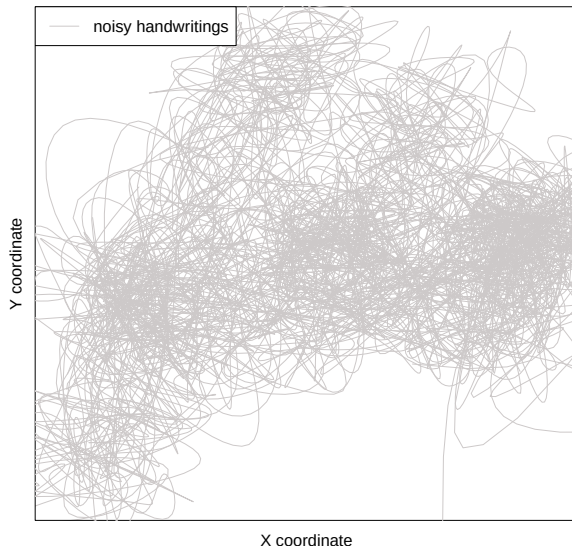
Motivating Example: Cursive Handwriting

noisy handwriting coordinates vs. time

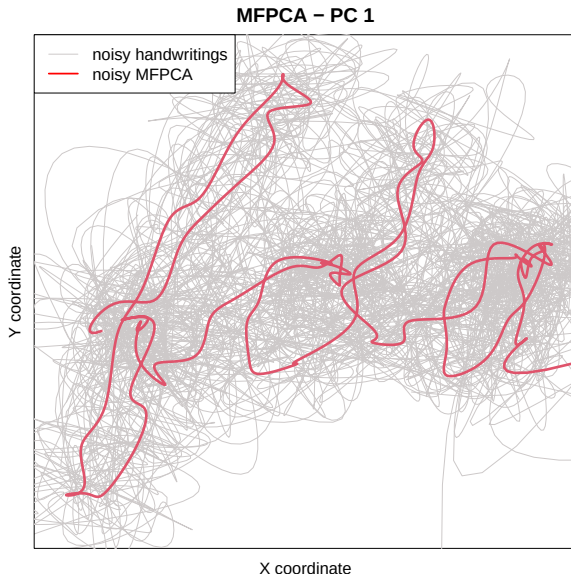


Motivating Example: Cursive Handwriting

fda – noisy handwriting samples



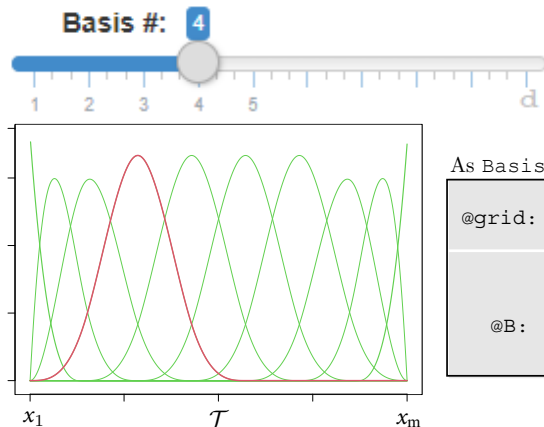
Motivating Example: Cursive Handwriting



Motivating Example: Cursive Handwriting

Motivating Example: Cursive Handwriting

Representation of the functional data (FD) in ReMFPCA

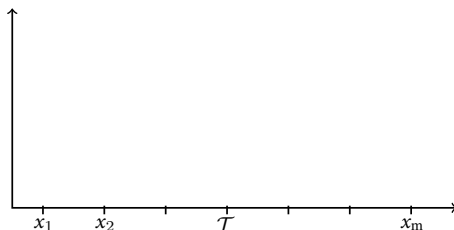


As Basis object:

@grid: vector $(x_1, x_2, \dots, x_m)$

@B: matrix $\begin{pmatrix} b_{11} & \dots & b_{d1} \\ \vdots & \dots & \vdots \\ b_{1m} & \dots & b_{dm} \end{pmatrix}$

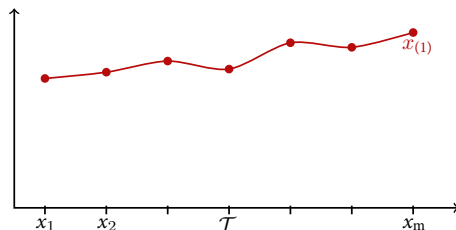
Representation of the functional data (FD) in ReMFPCA



As fd object:

@grid:	vector	$(x_1, x_2, \dots, x_m)$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{d1} \\ \vdots & \dots & \vdots \\ b_{1m} & \dots & b_{dm} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$

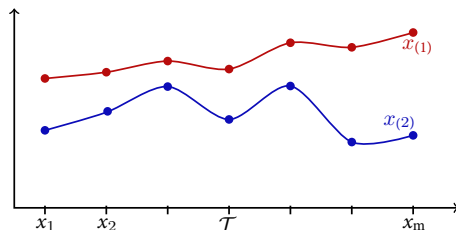
Representation of the functional data (FD) in ReMFPCA



As fd object:

@grid:	vector	$(x_1, x_2, \dots, x_m)$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{d1} \\ \vdots & \dots & \vdots \\ b_{1m} & \dots & b_{dm} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} c_{11} \\ \vdots \\ c_{1d} \end{pmatrix}$

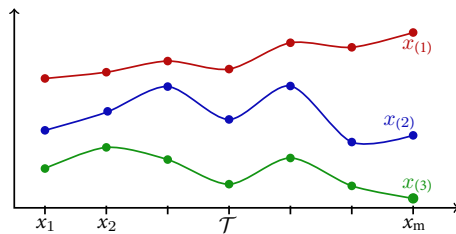
Representation of the functional data (FD) in ReMFPCA



As fd object:

@grid:	vector	$(x_1, x_2, \dots, x_m)$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{d1} \\ \vdots & \dots & \vdots \\ b_{1m} & \dots & b_{dm} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} c_{11} & c_{21} \\ \vdots & \vdots \\ c_{1d} & c_{2d} \end{pmatrix}$

Representation of the functional data (FD) in ReMFPCA



As fd object:

@grid:	vector	$(x_1, x_2, \dots, x_m)$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{d1} \\ \vdots & \dots & \vdots \\ b_{1m} & \dots & b_{dm} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} c_{11} & c_{21} & c_{31} & \dots \\ \vdots & \vdots & \vdots & \vdots \\ c_{1d} & c_{2d} & c_{3d} & \dots \end{pmatrix}$

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Representation of the functional data (FD) in ReMFPCA

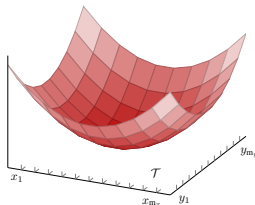
As fd object:

@grid:	matrix	$\begin{pmatrix} x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x} \\ y_1, y_1, \dots, y_1, y_2, y_2, \dots, y_2, \dots, y_{m_y} \end{pmatrix}$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{dx,1} & \dots & b_{d1} \\ \vdots & \dots & \vdots & \dots & \vdots \\ b_{1\tilde{m}} & \dots & b_{dx,\tilde{m}} & \dots & b_{d\tilde{m}} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \end{pmatrix}$

Representation of the functional data (FD) in ReMFPCA

As fd object:

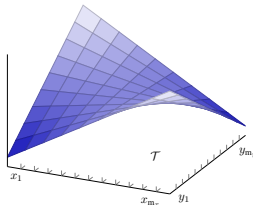
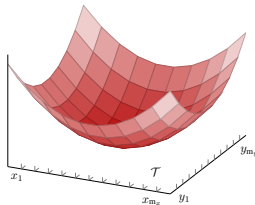
@grid:	matrix	$\begin{pmatrix} x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x} \\ y_1, y_1, \dots, y_1, y_2, y_2, \dots, y_2, \dots, y_{m_y} \end{pmatrix}$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{dx,1} & \dots & b_{d1} \\ \vdots & \dots & \vdots & \dots & \vdots \\ b_{1\tilde{m}} & \dots & b_{dx,\tilde{m}} & \dots & b_{d\tilde{m}} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} c_{11} \\ \vdots \\ c_{1d} \end{pmatrix}$



Representation of the functional data (FD) in ReMFPCA

As fd object:

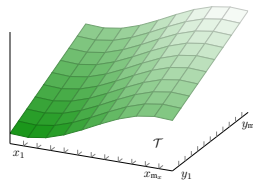
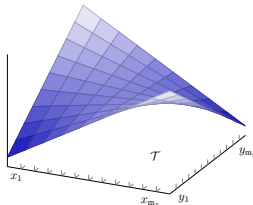
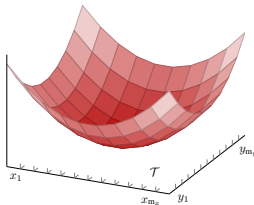
@grid:	matrix	$(x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x})$ $(y_1, y_1, \dots, y_1, y_2, y_2, \dots, y_2, \dots, y_{m_y})$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{dx,1} & \dots & b_{d1} \\ \vdots & \dots & \vdots & \dots & \vdots \\ b_{1\tilde{m}} & \dots & b_{dx,\tilde{m}} & \dots & b_{d\tilde{m}} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} c_{11} & c_{21} \\ \vdots & \vdots \\ c_{1d} & c_{2d} \end{pmatrix}$



Representation of the functional data (FD) in ReMFPCA

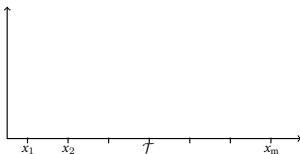
As fd object:

@grid:	matrix	$(x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x}, x_1, x_2, \dots, x_{m_x})$ $(y_1, y_1, \dots, y_1, y_2, y_2, \dots, y_2, \dots, y_{m_y})$
@B:	matrix	$\begin{pmatrix} b_{11} & \dots & b_{dx,1} & \dots & b_{d1} \\ \vdots & \dots & \vdots & \dots & \vdots \\ b_{1\tilde{m}} & \dots & b_{dx,\tilde{m}} & \dots & b_{d\tilde{m}} \end{pmatrix}$
@C:	matrix	$\begin{pmatrix} c_{11} & c_{21} & c_{31} & \dots \\ \vdots & \vdots & \vdots & \vdots \\ c_{1d} & c_{2d} & c_{3d} & \dots \end{pmatrix}$

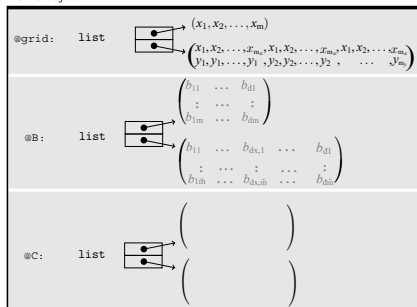


Representation of the functional data (FD) in ReMFPCA

- How about Multivariate Functional Data (MFD) observed over different dimensional domains?

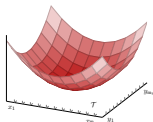
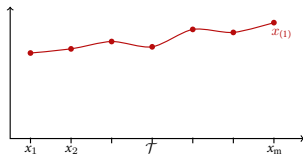


As fd object:

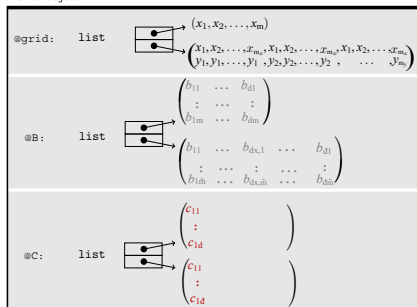


Representation of the functional data (FD) in ReMFPCA

- How about Multivariate Functional Data (MFD) observed over different dimensional domains?

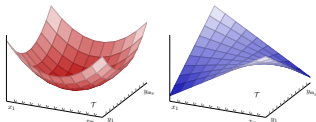
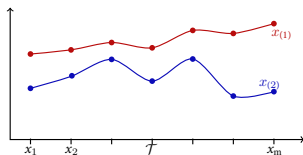


As fd object:

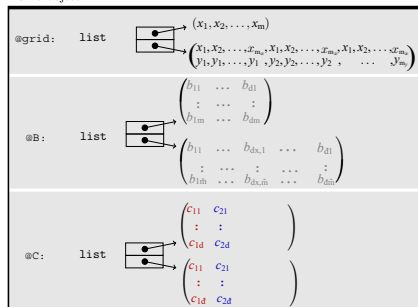


Representation of the functional data (FD) in ReMFPCA

- How about Multivariate Functional Data (MFD) observed over different dimensional domains?

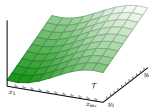
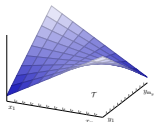
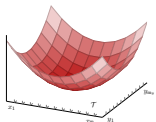
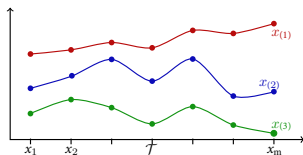


As fd object:



Representation of the functional data (FD) in ReMFPCA

- How about Multivariate Functional Data (MFD) observed over different dimensional domains?



As fd object:

